Which Atoms Need to Talk? Sparse, Transferable, Long-Range Edges for GNN Force Fields

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1 Motivation

Molecular dynamics (MD) simulates the movement of molecules by iteratively computing the force on each atom, then using those forces to calculate movement over very short time scales (typically, femtoseconds). These simulations are important for both structural biology and drug discovery. Classical force fields compute forces quickly but approximately from fixed functional forms. Alternatively, quantum methods like density functional theory (DFT), are more accurate but scale roughly cubically with system size, ruling out long simulations of complex biomolecules.

Machine-learned force fields (MLFFs) attempt to provide accurate MD simulations at lower computational costs. MLFFs are trained on DFT data and can reach near-DFT performance at a fraction of the cost [Unke et al., 2021]. The majority of MLFFs are GNNs in which nodes represent atoms and, for some fixed radius r_c, edges connect atoms less than r_c apart. Models predict the total energy E, and forces follow using $\mathbf{F}_i = -\partial E/\partial \mathbf{r}_i$.

The fixed radius cutoff keeps cost linear in system size, but distant atoms can only interact through several rounds of message passing, which dilutes their information. This hurts the non-covalent interactions that govern biomolecules, such as hydrogen bonds and π-stacking. Connecting all pairs of atoms fixes this but costs $O(N^2)$ and does not scale to proteins, while hand-written chemical rules for interacting pairs are incomplete. Our key observation is that whether two distant atoms interact depends mostly on their local chemistry and their distance, so it can be learned on small systems and applied to large ones. We learn this rule with a gate on each candidate edge and test whether it is accurate, efficient, chemically sensible, and transferable.

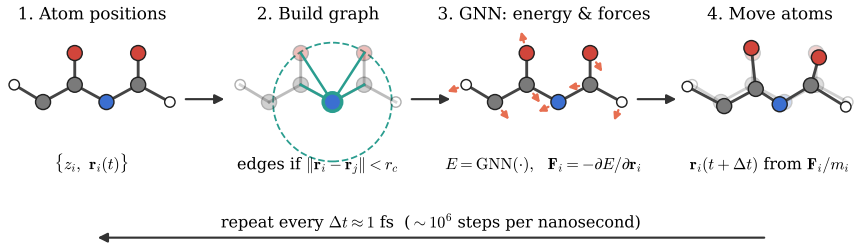


Figure 1: MD with a GNN force field using a fixed cutoff radius. At every step, the GNN computes the overall energy and forces acting on each atom. Atoms are moved according to Newton's laws.

2 Dataset

We use MD22 [Chmiela et al., 2023], a public benchmark (`http://sgdml.org`) of seven molecular systems with 42 to 370 atoms. We chose it because it was designed to include the non-local interactions our project studies. Each system is a set of configurations from *ab initio* MD at 400–500 K, labeled with DFT energies and forces. The model input is the atomic numbers and positions of a configuration, and the targets are its total energy and per-atom forces. We train one model per system.

We focus on two systems. AT-AT (60 atoms, 20,001 configurations) consists of two stacked adenine–thymine base pairs. This system is a clean test case because its four bases are not covalently bonded to each other but are held together only by hydrogen bonds and π-stacking, so every interaction between them is non-local and known in advance. Ac-Ala3-NHMe (42 atoms, 85,099 configurations) is a small peptide whose backbone hydrogen bonds form and break as its shape changes. In

a transfer experiment, we will also use AT-AT-CG-CG (118 atoms, 10,153 configurations), which adds two guanine–cytosine pairs.

3 Method

We implement a SchNet-inspired [Schütt et al., 2017] layer as a PyG `MessagePassing` module. It uses only interatomic distances, so the energy is invariant to rotations and translations. Node features $\mathbf{h}_i^{(0)}$ embed the atomic number z_i, and each layer updates

$$\mathbf{h}_i \leftarrow \mathbf{h}_i + \sum_{j:\, d_{ij}<r_c} \phi\big(\mathbf{h}_j, \mathrm{rbf}(d_{ij})\big)\, f_c(d_{ij}) \;+ \sum_{j:\, r_c \le d_{ij}<r_\ell} g_{ij}\, \tilde{\phi}\big(\mathbf{h}_j, \mathrm{rbf}(d_{ij})\big), \tag{1}$$

where d_{ij} is the interatomic distance, rbf is a radial basis expansion (a standard encoding of distance, easier to learn from than the raw scalar), and f_c is a smooth envelope that vanishes at the short cutoff r_c. The first sum is a standard cutoff model. The second sum is our contribution. It runs over candidate long-range edges up to $r_\ell > r_c$, each scaled by a learned gate g_{ij}. The energy is $E = \sum_i \psi(\mathbf{h}_i^{(L)})$ and forces are $\mathbf{F}_i = -\partial E/\partial \mathbf{r}_i$ via autograd.

Our main contribution is the **learned gate**. Each candidate edge receives

$$g_{ij} = \sigma\big(\mathrm{MLP}(\mathbf{h}_i + \mathbf{h}_j,\ \mathbf{h}_i \odot \mathbf{h}_j,\ \mathrm{rbf}(d_{ij}))\big)\, w(d_{ij}), \tag{2}$$

computed once per configuration from embeddings after a first, short-range-only layer, so the gate sees local chemistry. The inputs are symmetric, so $g_{ij} = g_{ji}$, and long-range messages $\tilde{\phi}$ are normalized so that a small gate really means a weak edge. The window w is smooth and vanishes at r_c and r_ℓ. Gates are continuous rather than a hard choice of edges because forces are derivatives of the energy, and any jump in the energy would break energy conservation in MD. We train with

$$\mathcal{L} = \underbrace{w_E(\hat{E} - E)^2}_{\text{energy error}} + \underbrace{\tfrac{w_F}{N}\textstyle\sum_i \|\hat{\mathbf{F}}_i - \mathbf{F}_i\|^2}_{\text{force error}} + \underbrace{\lambda \cdot \mathrm{mean}(g_{ij})}_{\text{sparsity penalty}}, \tag{3}$$

where w_E and w_F balance the two error terms. The sparsity penalty keeps a gate open only if it improves accuracy enough to pay its cost, and we ramp λ up from zero to avoid closing all gates early. Because the gate depends on local environments and distance rather than a fixed edge list, a gate trained on one system can be applied to a larger one. This suits MD22, whose systems contain non-local interactions a short cutoff misses, and AT-AT, whose true inter-base contacts are known.

We compare the gated model with two alternatives on the same backbone. The *dense* model keeps all candidate edges with $g_{ij} = 1$. It is the most complete long-range model we can build, but its cost grows rapidly with r_ℓ and approaches all N^2 pairs for our systems, so it tests whether a sparse learned graph gives up accuracy. The *chemistry-typed* model keeps only long-range pairs chosen by hand-written rules, namely hydrogen-bond donor–acceptor pairs and pairs of aromatic ring atoms, with a separate message function per type as in relational GCNs [Schlichtkrull et al., 2018]. Like the gate, this rule applies to any molecule, so it is the natural competitor in transfer. A cutoff-only model serves as a lower bound.

4 Experiments and analysis

Accuracy versus sparsity: We sweep λ and plot force and energy mean absolute error (MAE) against the fraction of candidate edges kept, for AT-AT and the peptide, with the dense, typed, and cutoff-only models for reference. We report three seeds and use the same splits as published MD22 results [Kovács et al., 2023] and compare to the results in the paper. The key question is how few edges recover the accuracy of the dense model.

Data efficiency: We train the gated and dense models on increasing numbers of configurations and compare learning curves. A sparse graph is a stronger inductive bias, so we expect the gated model to reach a given accuracy with less data.

Transfer to a larger nucleic acid: We apply the gate function trained on AT-AT to AT-AT-CG-CG, which contains guanine and cytosine bases the model has never seen. We first check whether the predicted gates pick out the correct contacts without retraining. We then train force fields on AT-AT-CG-CG with the transferred gates frozen and compare them with gates learned from scratch and with the dense and typed models, in both accuracy and number of edges.

Simulation: Even small errors in forces add up over millions of steps. We will run MD simulations with our learned force fields and see how soon they blow up.

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